Supplementary Derivations

Exact Logical-Channel Equivalence and Code-Space-Leakage-Aware Tomography of a Five-Qubit Encoded-Qubit Transfer Circuit

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This supplement expands the compact proof in Section 3 and Appendix A of the revised article. The operator identity applies to the specified ideal circuit; it does not assert exact noisy-device performance. Tensor factors are written in descending wire order. The signs and binary labels have the same meanings as in the main text.

S1 Tensor-embedded controlled-X notation

For an n-qubit register $|b_{n-1}\cdots b_0\rangle$, let

$$O_{[j]} = I_{n-1}\otimes\cdots\otimes O_j\otimes\cdots\otimes I_0.$$

For distinct control i and target j,

$$\mathrm{CX}_{i,j,n} = \big(|0\rangle\langle 0|\big)_{[i]} + \big(|1\rangle\langle 1|\big)_{[i]} X_{[j]}. \tag{S1}$$

The implicit identity on the target in the first term, and on every other factor in both terms, makes the embedding explicit. Equivalently, it maps b_j to $b_j \oplus b_i$ and leaves every other bit unchanged. This is standard controlled-gate algebra; the extended Dirac notation is related to Yamanashi, Ozawa and Okazaki, IEICE Technical Report 122, CAS2022-61, 1–6 (2023), cited in the main article. No new controlled-gate formalism is claimed.

S2 Gate-by-gate logical-basis derivation

Fix $x, y, m, M_1, M_2 \in \{0,1\}$, write $z = m \oplus x$ and $\kappa = m \oplus M_1 \oplus M_2$, and define

$$|\lambda_L^{(y)}\rangle = |y\oplus\lambda, \lambda\rangle\,, \qquad |G_{m;M_1,M_2}\rangle = \frac{1}{\sqrt{2}}\sum_{\eta=0}^{1}(-1)^{m\eta}\,|M_2\oplus\eta, M_1\oplus\eta, \eta\rangle\,. \tag{S2}$$

Here λ is the logical basis bit and η is the resource branch bit. The five Hadamard summation bits below are denoted $a_1, \ldots, a_5$ and are local dummy variables. Binary exponents of (-1) are evaluated modulo two; the written sums run over each indicated bit in $\{0,1\}$.

The initial five-qubit state is

$$|\Psi_\lambda^{(0)}\rangle = \frac{1}{\sqrt{2}}\sum_{\eta}(-1)^{m\eta}\,|M_2\oplus\eta, M_1\oplus\eta, \eta, y\oplus\lambda, \lambda\rangle_{43210}\,. \tag{S3}$$

Read the main circuit from left to right. The first four gates $H_1, \mathrm{CX}_{1,2,5}, H_1, \mathrm{CX}_{2,3,5}$ give

$$|\Psi_\lambda^{(1)}\rangle = \frac{1}{2\sqrt{2}}\sum_{\eta,a_1,a_2}(-1)^{m\eta+a_1(y\oplus\lambda)+a_1a_2}\,|M_2\oplus\eta, M_1\oplus a_1, \eta\oplus a_1, a_2, \lambda\rangle\,. \tag{S4}$$

The next block $H_3, X_3^m, \mathrm{CX}_{1,3,5}, H_3, X_3^m$ yields

$$|\Psi_\lambda^{(2)}\rangle = \frac{1}{4\sqrt{2}}\sum_{\eta,a_1,a_2,a_3,a_4}(-1)^{P_2}\,|M_2\oplus\eta, a_4\oplus m, \eta\oplus a_1, a_2, \lambda\rangle\,, \tag{S5}$$

$$P_2 = m\eta + a_1(y\oplus\lambda) + a_1a_2 + a_3(M_1\oplus a_1) + a_4(a_3\oplus m\oplus a_2). \tag{S6}$$

The three controlled gates $\mathrm{CX}_{2,4,5}, \mathrm{CX}_{4,3,5}, \mathrm{CX}_{0,3,5}$ change only the basis labels to

$$|M_2\oplus a_1, a_4\oplus m\oplus M_2\oplus a_1\oplus\lambda, \eta\oplus a_1, a_2, \lambda\rangle\,. \tag{S7}$$

Finally, $H_4, X_4^z, \mathrm{CX}_{4,0,5}$ give

$$|\Psi_\lambda^{(f)}\rangle = \frac{1}{8} \sum_{\eta,a_1,\ldots,a_5} (-1)^P \, |a_5 \oplus z, a_4 \oplus m \oplus M_2 \oplus a_1 \oplus \lambda, \eta \oplus a_1, a_2, \lambda \oplus a_5 \oplus z\rangle \,, \tag{S8}$$

$$P = P_2 + a_5(M_2 \oplus a_1). \tag{S9}$$

These expressions follow directly from $H\,|b\rangle = 2^{-1/2}\sum_a (-1)^{ba}\,|a\rangle$ and $X^c\,|b\rangle = |b \oplus c\rangle$.

Set $\zeta = \eta \oplus a_1$. The sum over a_3 enforces $a_4 = M_1 \oplus a_1$. The remaining exponent becomes

$$P = m\zeta + mM_1 + M_1 a_2 + a_5 M_2 + a_1(y \oplus \lambda \oplus a_5). \tag{S10}$$

The sum over a_1 therefore enforces $a_5 = y \oplus \lambda$. The final labels are

$$(q_{4'}, q_{3'}, q_{2'}, q_{1'}, q_{0'}) = (y \oplus \lambda \oplus z, \lambda \oplus \kappa, \zeta, a_2, y \oplus z). \tag{S11}$$

The remaining ζ, a_2 sums are $H\,|m\rangle_{2'}$ and $H\,|M_1\rangle_{1'}$. Thus

$$\begin{aligned} U_T^{(x,m)}\big(|G_{m;M_1,M_2}\rangle \otimes |\lambda_L^{(y)}\rangle\big) = (-1)^{mM_1 + M_2(y\oplus\lambda)}\, &|y \oplus \lambda \oplus z, \lambda \oplus \kappa\rangle_{4'3'} \\ &\otimes H\,|m\rangle_{2'} \otimes H\,|M_1\rangle_{1'} \otimes |m \oplus x \oplus y\rangle_{0'} \,. \end{aligned} \tag{S12}$$

The discarded vector is independent of λ. The only λ-dependent phase is $(-1)^{M_2\lambda}$, which is exactly the Z^{M_2} factor of

$$C = X^z \otimes X^\kappa Z^{M_2}, \qquad C^\dagger = X^z \otimes Z^{M_2} X^\kappa, \qquad C^\dagger C = I. \tag{S13}$$

The inverse is an exact operator identity, with no unspecified phase. Linearity yields the same receiver map for every encoded matrix unit $|i_L\rangle\langle j_L|$, including $i \neq j$, and hence for arbitrary mixed states and reference-entangled inputs supported on the fixed code.

S3 Why two physical input qubits are not two logical qubits

For a fixed parity y, $V_y\,|\lambda\rangle = |y \oplus \lambda, \lambda\rangle$ has a two-dimensional image. Let $P_y = V_y V_y^\dagger$. The discarded output $|m \oplus x \oplus y\rangle_{0'}$ distinguishes $y = 0$ from $y = 1$. Therefore, after correction, tracing out discarded wires removes the cross-parity blocks:

$$\mathcal{R} \circ \mathcal{T}(\rho) = P_0 \rho P_0 + P_1 \rho P_1. \tag{S14}$$

The ordered input/output identification is understood. Possible parity-dependent global signs in the isometries cancel within each diagonal block and cannot restore the discarded cross-parity coherence. This proves identity only on either fixed code, not on the unrestricted four-dimensional input space. The physical measurements in run02 use $y = 0$; the statement for $y = 1$ is theoretical.

The supplied audit independently checks 64 basis-state instances, 128 fixed-code matrix-unit instances, and 256 unrestricted-input matrix-unit instances against the released ideal unitary. Maximum absolute discrepancies are below 10^{-15}. These are numerical checks of the stated algebra, not extra hardware measurements.